

Harry Lian

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EDUCATION

Ohio State University

Columbus, OH

Bachelor of Science Electrical & Computer Engineering (Computer Engineering)

May 2026

Awards/Honors: Dean's List, Awarded 3rd in Engineering Showcase, Daytona LaSertoma Recipient

Relevant Coursework: Data Structures and Algorithms, Machine Learning, Operating Systems, Computer Architecture

WORK EXPERIENCE

Ohio State University

Columbus, OH

Electronic Devices Lab Monitor

Sept. 2025 – May 2026

- Diagnosed and resolved faults in analog and digital circuits across 3 weekly lab sections supporting 20+ students each, reducing average troubleshooting time by 35% through systematic isolation methodologies
- Guided 60+ students per week through LTspice simulation and oscilloscope measurement workflows, improving average lab report accuracy by 20% through structured pre-lab walkthroughs and real-time fault diagnosis

Reality AI Lab

Remote

Software Developer Intern

May 2025 – Sept. 2025

- Reduced average page load time by 30% across 5+ React interfaces by implementing component-level code splitting, lazy loading, and memoization, backed by RESTful APIs in Node.js and Python
- Cut mean time to bug resolution from 4 hours to under 90 minutes by owning end-to-end debugging workflows using LogRocket session replay and Log Explorer, contributing to a 40% reduction in production bug rate over the internship period
- Engineered core retrieval and ranking logic for a real-time AI recommendation engine - built in Python with vector similarity search and integrated into a Node.js microservice - serving 10,000+ daily active users

PROJECTS

Haptic Feedback VR Glove - Senior Capstone | C++, Arduino, ESP32, BLE, PWM, Unity

- Engineered BLE firmware on dual ESP32-S3 microcontrollers - server paired to Unity, client embedded in glove - implementing bidirectional communication with ACK/status feedback and automatic reconnection on disconnect
- Designed a PWM actuation system controlling 10 vibration motors and 2 Peltier thermal tiles at 3 intensity levels (low/med/high) via ledcWrite, with a 15-second hardware safety timer on Peltier tiles to prevent overheating
- Reduced Unity integration latency by building a custom multi-command BLE packet parser supporting simultaneous haptic and thermal control in a single transmission, eliminating the need for multiple roundtrips and simplifying game-side logic

Autonomous Oxygen Management System - IEEE ProfHackathon (1st Place) - Python, Modbus, PLC, VFD

- Won 1st place in Track 3 (Outlaws & Automation) by designing and building a fully autonomous simulation: a Python-controlled PLC acting as a space habitat oxygen tank that dynamically responds to demand fluctuations without manual input
- Implemented a real-time Serial/Modbus control loop in Python to autonomously poll O2 demand, adjust motor output via Variable Frequency Drive, and update a live concentration display

RAG API with Automated Quality Testing - Python, FastAPI

- Built a FastAPI REST API implementing Retrieval-Augmented Generation: queries ChromaDB for nearest-neighbor context retrieval, augments prompts, and generates responses via a local Ollama (tinylama) LLM - all within a single /query endpoint
- Prevented data quality regressions from reaching production by implementing a GitHub Actions CI/CD pipeline that runs automated semantic regression tests on every push against a multi-document ChromaDB knowledge base

TECHNICAL SKILLS

- Languages: Python, Java, SQL, C++, C, JavaScript, HTML/CSS, Mandarin (Fluent)
- Databases & Data: ChromaDB, PostgreSQL, vector databases
- Cloud & DevOps: AWS (S3, EC2, Lambda), Docker, Kubernetes, Linux, CI/CD
- Frameworks & Libraries: React, Node.js, FastAPI, NumPy, Pandas